CS411W Lab

Take The Lead

<M. Lacey, Red Team>

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to document the requirements and expected behavior of the Take The Lead (TTL) software project. This document provides a structured reference for the development team to understand the system's intended functionality, architecture, user roles, constraints, and dependencies. It also establishes a basis for evaluating the prototype against the project's goals and for guiding future development.

1.2 Scope

Take The Lead (TTL) is a proposed web application designed to help pet owners obtain timely veterinary care when unexpected expenses or financial hardship make necessary treatment difficult to afford. The application aims to support pet owners through fundraising campaigns, promise-to-pay arrangements, and access to educational resources related to veterinary care and financial assistance.

The Take The Lead application will allow registered users, referred to as Members, to create campaigns describing their pets' veterinary needs. Each campaign will include a 48-hour preparation period followed by a 30-day fundraising period. The intended promise-to-pay arrangement will be an agreement between the Member and Take The Lead prior to the start of the fundraising period under which TTL will pay the veterinary bill up to an agreed amount, with a proposed maximum of $5,000. The Member will be required to repay TTL before funds are

distributed to the veterinary hospital. Veterinary costs will be verified by TTL through direct contact with the hospital.

In addition to fundraising, Take The Lead will provide, in future rollouts, Members and Guests with educational resources about veterinary services, preventative care, pet insurance, pet-specific credit options, and the challenges faced by veterinary professionals. The application is also intended to provide a directory of emergency and urgent-care veterinary services organized by ZIP code, information about emergency contacts and Poison Control, and an AI-powered chatbot to help users navigate the application and locate relevant information. The chatbot may also assist Members with improving the grammar and clarity of campaign Stories, but it will not create campaigns or provide veterinary diagnoses.

The prototype will demonstrate selected application workflows rather than provide a complete financial assistance service. Campaign data will be stored in a database, and time-based campaign logic will be implemented. Donations will be simulated, with donation totals and campaign balances displayed; actual payments, promise-to-pay obligations, and final fund distribution will not be implemented. The prototype will use illustrative veterinary directory entries and a sample dataset. An external AI service or API is planned for the chatbot, although the provider has not yet been selected.

Through these features, Take The Lead aims to improve access to veterinary care, help pet owners prepare for unexpected expenses, encourage informed decisions about animal health, and provide greater awareness of the challenges experienced by veterinary professionals.

1.3 Definitions, Acronyms, and Abbreviations

Campaign: A fundraising effort containing a Member's Story and its associated content and funds.

Compassion Fatigue/Compassion Burnout: Profound physical, emotional, and psychological exhaustion caused by repeated and prolonged exposure to trauma and suffering.

ER (Emergency Room): A veterinary emergency facility that provides urgent treatment for animals, often operating 24 hours a day.

Guest: A nonregistered user who may view campaign Stories, contribute donations, and access available informational resources.

MDP (Minimum Down Payment): An agreed monetary amount associated with a financial arrangement intended to allow a veterinary hospital to proceed with treatment. The proposed maximum amount for the Take The Lead arrangement is $5,000.

Member: A registered user of the Take The Lead web application who may create and manage a campaign. In a household with multiple users, the Member who owns a campaign is responsible for managing its Story.

MFCD (Major Functional Component Diagram): A visual representation of an application's primary functional components and the relationships and information flows between them.

NOMV (Not One More Vet): An organization focused on veterinary mental health, well-being, and suicide prevention.

Preventative Care: Animal care intended to prevent illness or identify potential health problems before they become more serious. Examples include vaccinations, routine examinations, and wellness checkups.

Promise-to-Pay Agreement: An agreement between a Member and Take The Lead establishing the amount TTL will pay toward a veterinary bill and the Member's obligation to repay TTL before funds are distributed to the veterinary hospital.

Story: The written content, photographs, videos, and other information a Member includes in a campaign to explain the circumstances surrounding a pet's need for veterinary assistance.

Software Requirements Specification (SRS): a formal document that describe es what a software system must do, how it should behave, and what constraints it must operate under acting as a single source of truth between stakeholders, designers, and developers.

TTL (Take The Lead): The proposed web application and service intended to help pet owners access veterinary financial assistance, fundraising, and educational resources.

UC (Urgent Care): A veterinary facility that provides care for conditions requiring prompt attention but that may not require an emergency hospital. Availability, hours, and treatment capabilities vary by facility.

Vet: Informal shorthand for veterinarian.

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More information:

https://docs.google.com/presentation/d/1fMT1mr5lzd49JsqnMI66UK1awXY30v3f7sXNBwCdtjw/edit?slide=id.g3d86e310d8e_1_0#slide=id.g3d86e310d8e_1_0

Take The Lead Temporary Website: https://teach210.github.io/take-the-lead/

2. Overall Description

2.1 Product Perspective

Take The Lead (TTL) is a proposed web application that helps pet owners seek financial assistance for veterinary expenses. Members create campaigns, while Guests can browse campaigns and access available resources. Veterinary hospitals are external stakeholders involved in verifying veterinary expenses.

Figure 1

Major Functional Component Diagram



The MFCD illustrates the proposed architecture, consisting of a frontend, backend, database and external services. The frontend which provides the user interface, navigation, forms, and display of application data. Svelte and Tailwind CSS are identified as the proposed technologies. Backend/Application Layer will handle intended business logic, including authentication, campaign management, donations, fund distribution, and resource management. The database uses SQLite with Prisma ORM to store application data, including campaign information. The frontend communicates with the backend, which processes requests and interacts with the database or external services as needed. External services will have planned integrations to include payment processing, AI chatbot functionality, and other external APIs.

The frontend communicates with the backend, which processes requests and interacts with the database or external services as needed. The diagram represents the proposed architecture, not a claim that every component is operational.

The current prototype stores campaign data, implements campaign timing logic, and simulates donations. It displays donation totals and campaign balances but does not process real payments, distribute funds, or implement the intended promise-to-pay agreement and repayment process.

2.2 Product Functions

Table 1 summarizes the prototype's features compared with the intended real-world product.

Table 1

Take The Lead Prototype Feature Status

Feature	Prototype status
User accounts	Partial: Simple login; no MFA or account verification.
Pet Story creation	Partial: Basic text entry with limitations on text and photos.
Donation system	Partial: Simulated donations and displayed totals; no live payments or receipts.
Fund distribution	Not implemented: Design concept only; no funds are distributed.
Notifications	Not implemented: No email or SMS notifications.
Payment security	Not implemented: No real payment processing or associated payment security.
User Story verification	Manual procedure outside the application: A representative contacts the relevant office to confirm user details.

User/veterinarian verification	Not implemented**:** Automated verification and data comparison are unavailable.
Transparency	Partial: Static information and simulated balances; no real-time fund tracking.
Legal and compliance features	Not implemented: No legal, financial-compliance, or tax-documentation workflows.
Data storage	Partial: Basic storage; production cloud storage, backups, and encryption are not implemented as full capabilities.
Scalability	Partial: Limited demonstration; no production load balancing or traffic management.
User interface	Partial: Basic prototype interface rather than the planned polished, accessible production interface.
Customer support	Not implemented: No operational support system or confirmed AI chatbot integration.

Analytics	Not implemented: No analytics dashboard or full donation-tracking reports.

The prototype also demonstrates campaign timing, including the intended 48-hour preparation period and 30-day fundraising period. The real-world product is intended to add secure payments, fund distribution, promise-to-pay agreements, enhanced verification, notifications, customer support, and additional tracking capabilities.

2.3 User Characteristics

Members, or registered users who create and manage pet fundraising campaigns. They are expected to have basic web-browser and online-form experience. Guests or nonregistered users who browse campaigns and access available resources. They need basic web-browsing skills. Veterinary hospitals or external stakeholders who provide veterinary services and may be contacted to confirm expenses. They have no integrated portal in the prototype. Take The Lead Administrators or Representatives or Agents must coordinate manual verification and assistance-related activities. Full administrative and financial workflows are not implemented.

Verification of user-submitted Story information is performed manually outside the application by contacting the relevant office. This process is not an automated software feature.

All users are expected to have internet access and a compatible web browser. Available tasks are limited to the functionality implemented in the prototype.

2.4 Constraints

Donations are simulated; real payments, receipts, and fund distribution are unavailable. Promise-to-pay agreements and repayment are not implemented. Verification depends on manual contact where applicable; automated veterinary-system integration is unavailable. Authentication, notifications, customer support, analytics, and production-level security are incomplete or absent. Storage and scalability are limited compared with the intended production system. A real-world service would require appropriate legal, financial, privacy, and compliance review before deployment for the promise-to-pay and any future documents.

2.5 Assumptions and Dependencies

Users require an internet connection and compatible web browser. Campaign functionality depends on the available data storage and time-based logic. Manual verification depends on representatives contacting veterinary offices. A production system would require a payment provider, appropriate hosting and database deployment, backups, and security measures. The proposed chatbot would depend on an external AI service; the provider and integration have not been confirmed. External veterinary and pet-care information would depend on the availability and reliability of the relevant sources. The prototype's illustrative directory is not a live, verified directory.